

Introduction to Diagnostic Ultrasound Imaging

Brief history of Sonography

- Sound-based technologies were used for ships avoiding rocky shores (rudimentary Hydrophones)
- Hydrophones for inter-ship communication – Morse Code
- Then echo based detecting of fish (multi directional hydrophones) that were then repurposed for detecting submarines
- World war – I: for the detection of enemy submarines (detection of German U-boats) using echo detection
- Tuned hydrophones
- Piezo electric crystals for Ultrasound generation

Key concepts

- Pressure Wave

Pressure vs time, Pressure vs space

- Longitudinal wave velocity and wavelength
- Intensity & Power
- Attenuation and depth of Ultrasound penetration
- Reflection and Scattering
 - Acoustic impedance

$$z = \rho \times c$$

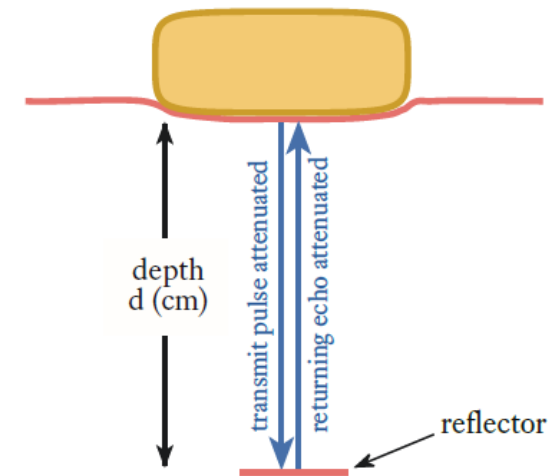
$$z = \frac{p}{v}$$

p is the local pressure
v is the local particle velocity in the medium
(this is different from wave velocity)

Acoustic impedance, z , of a medium is a measure of the response of the medium in terms of local particle velocity, to a wave of a given pressure.

$$\text{attenuation} = (\alpha \times L \times f) \text{ dB}$$

$$\text{attenuation} = (10 \log \frac{I_1}{I_2}) \text{ dB}$$

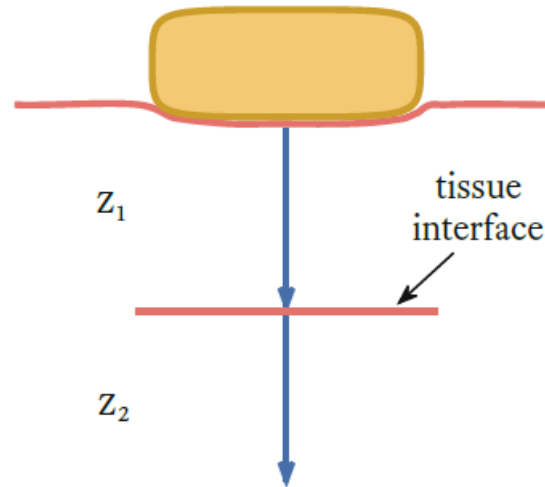
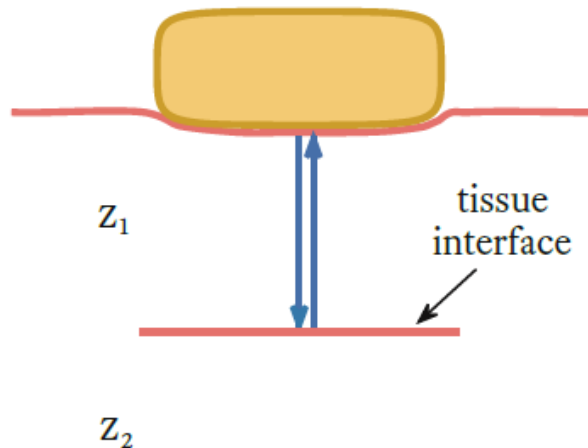


Analogous to $R=V/I$ for electric conduction

Acoustic impedance and reflectance

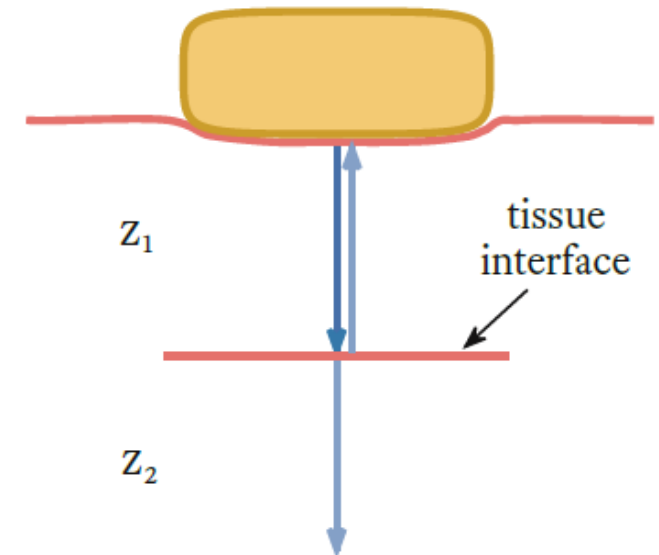
- Acoustic Impedance
- Proportion of energy reflected (reflectance)

$$R = \frac{\text{reflected intensity}}{\text{incident intensity}} \\ = \frac{(z_1 - z_2)^2}{(z_1 + z_2)^2}$$



Transmission coefficient T

$$T = \frac{\text{transmitted intensity}}{\text{incident intensity}} \\ = \frac{(4 \times z_1 \times z_2)}{(z_1 + z_2)^2}$$



Scattering

- Scattering 'reflects' the energy in all directions. This leads to *speckle* pattern in the image.

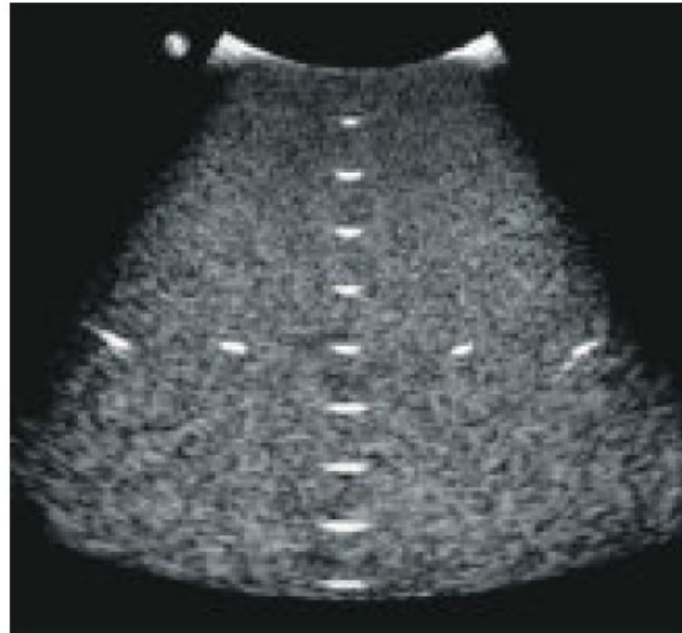
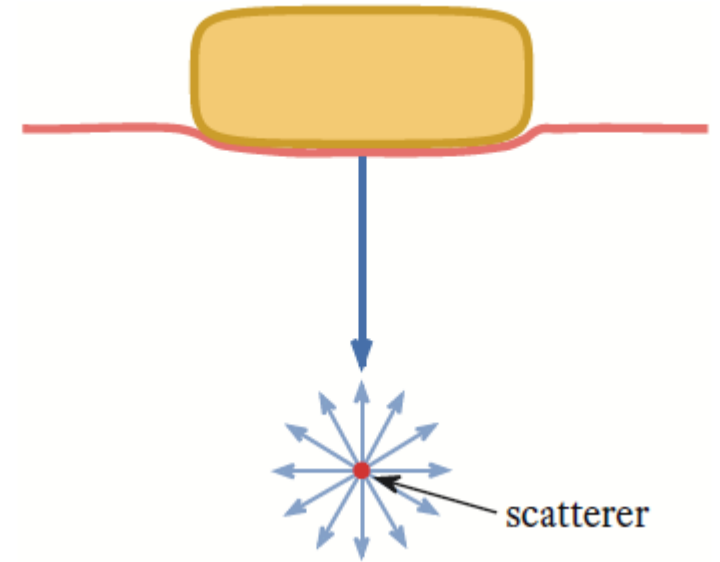


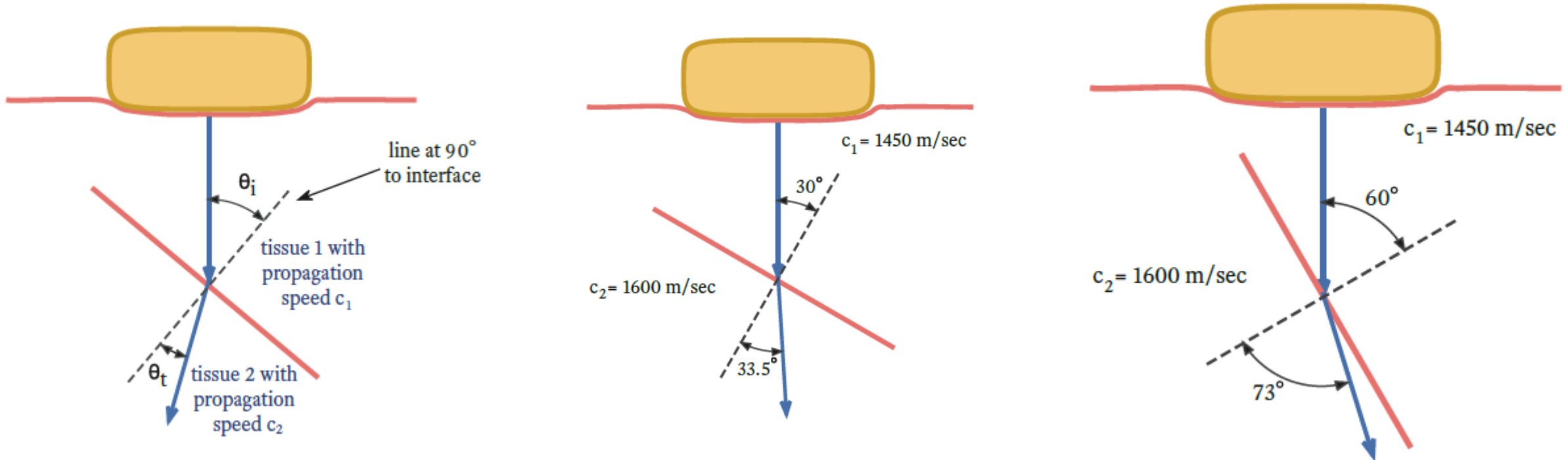
Figure 2.13 Scan of an ultrasound phantom (test object) showing speckle.



Refraction

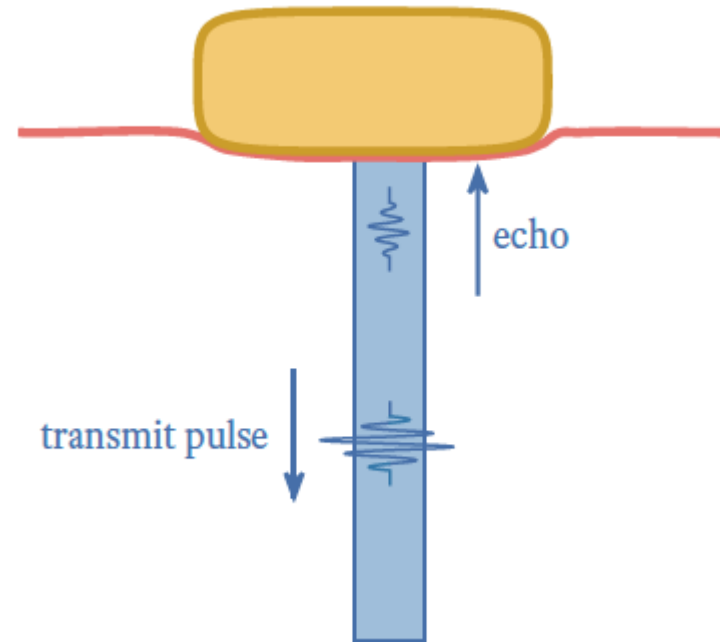
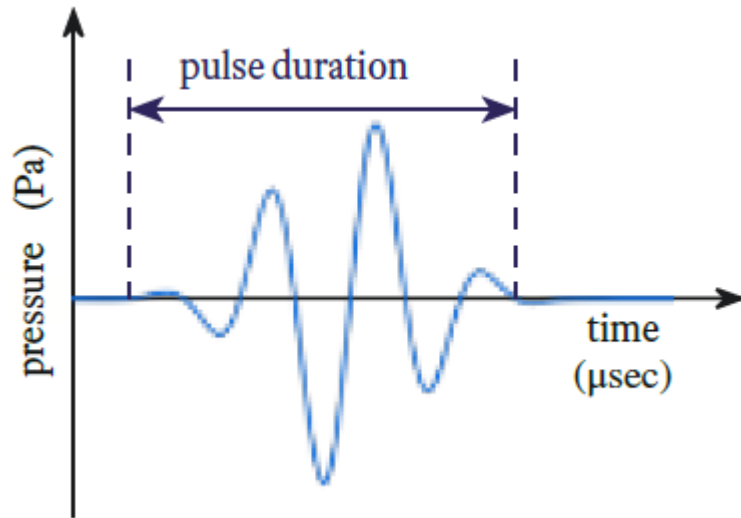
- Bending of the sound wave at tissue interfaces where the velocity of sound is significantly different.

$$\frac{\sin \theta_i}{c_1} = \frac{\sin \theta_t}{c_2}$$



Imaging with Pulsed Ultrasound

- Pulse width and pulse repetition frequency



$$d = \frac{c \times t}{2}$$

Imaging with Pulsed Ultrasound

- Pulse width and pulse repetition frequency

$$t_P = \frac{2 \times P}{c}$$

$$\text{maximum PRF} = \frac{c}{2 \times P}$$

A – Mode

A - for Amplitude

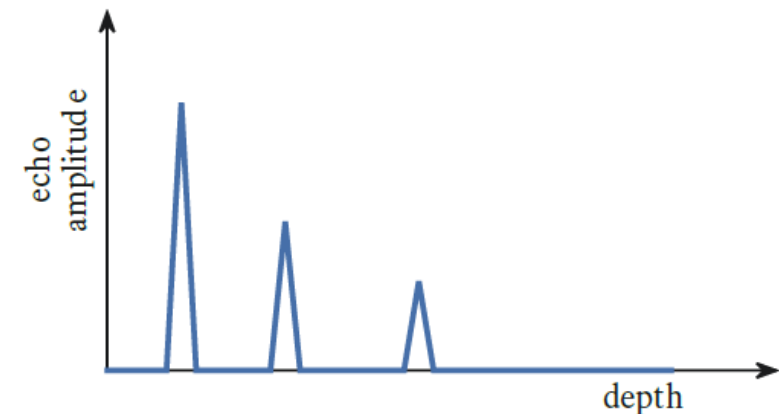
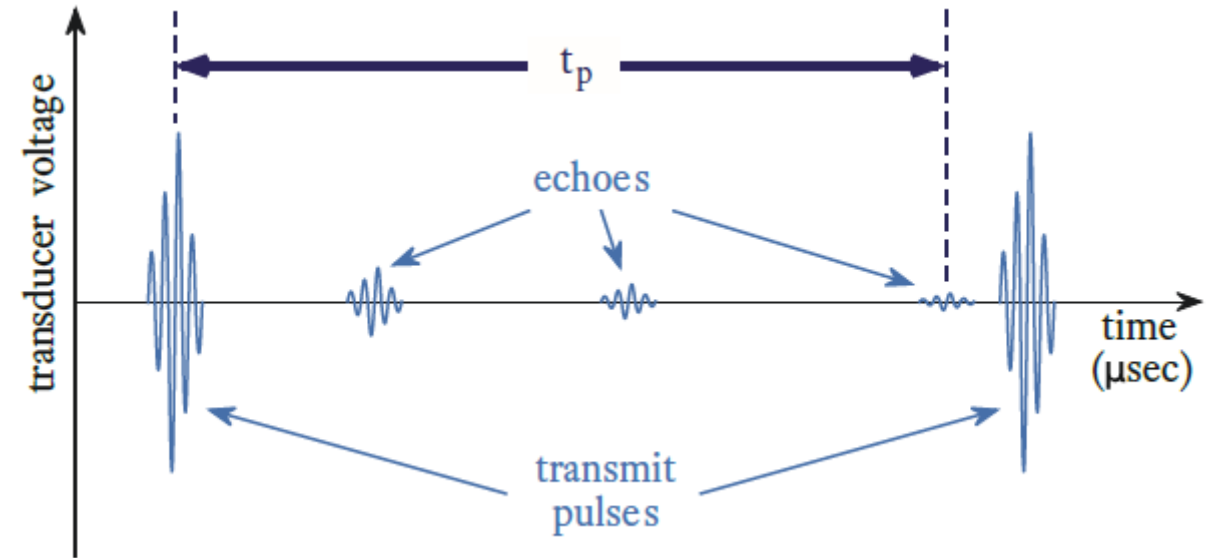


Figure 3.14 An A mode display.

Imaging with Pulsed Ultrasound

- Frame-rate
- N is the number of lines required for generating a B-mode image

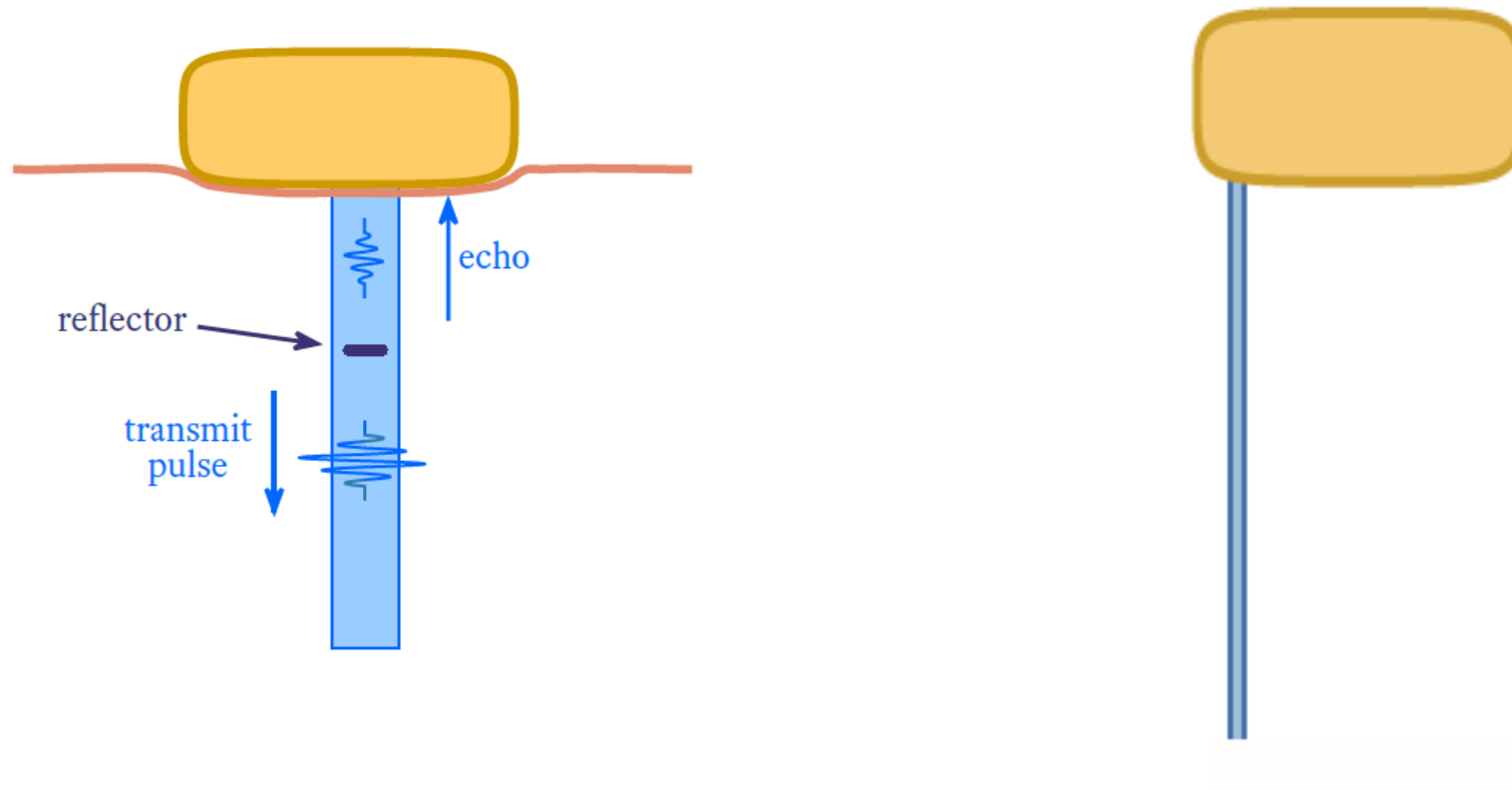
$$N \times FR = PRF$$

$$\text{maximum } FR = \frac{c}{2 \times P \times N}$$

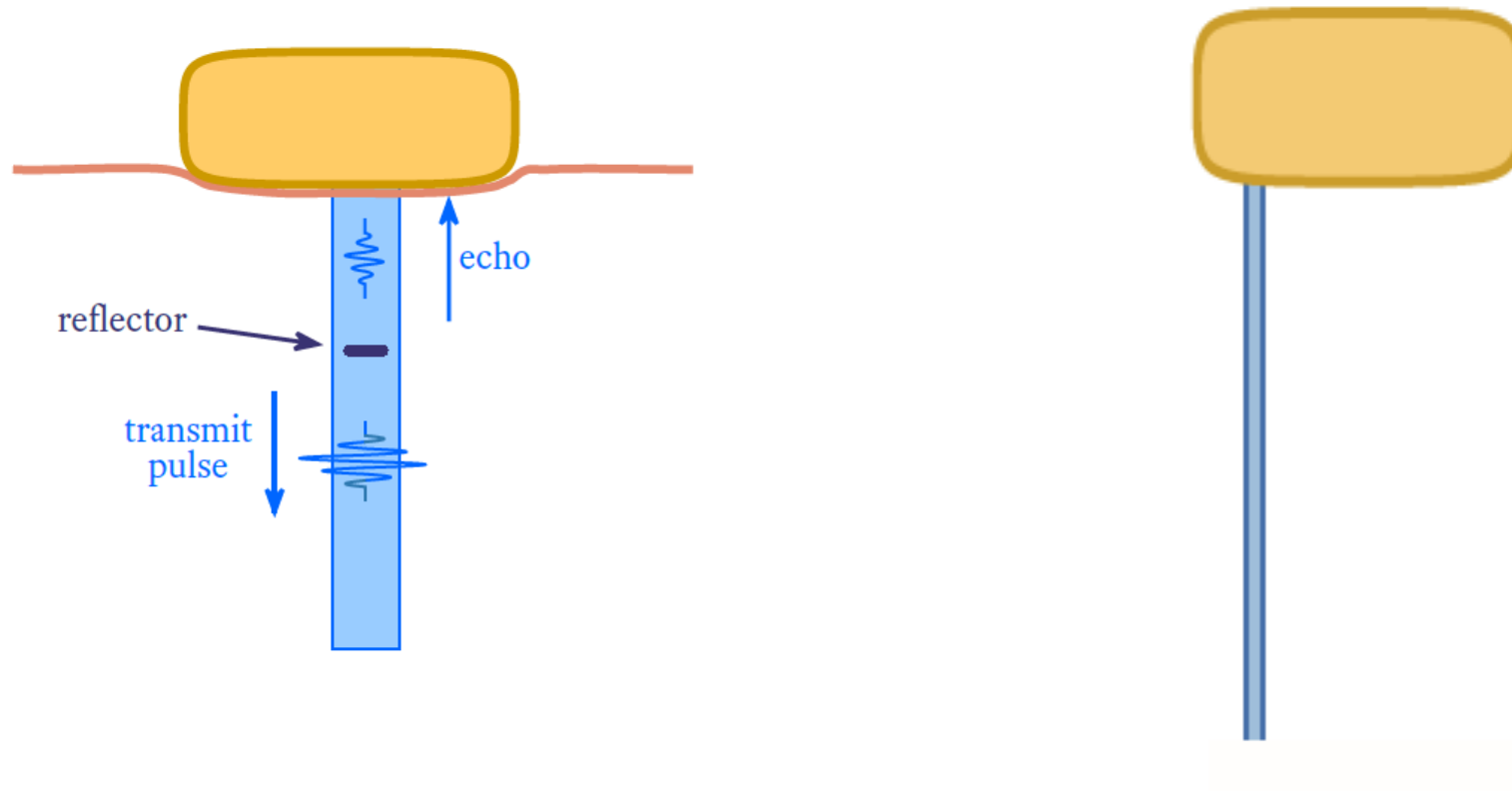
- FR here is frame rate possible

$$(FR \times P \times N) = \frac{c}{2}$$

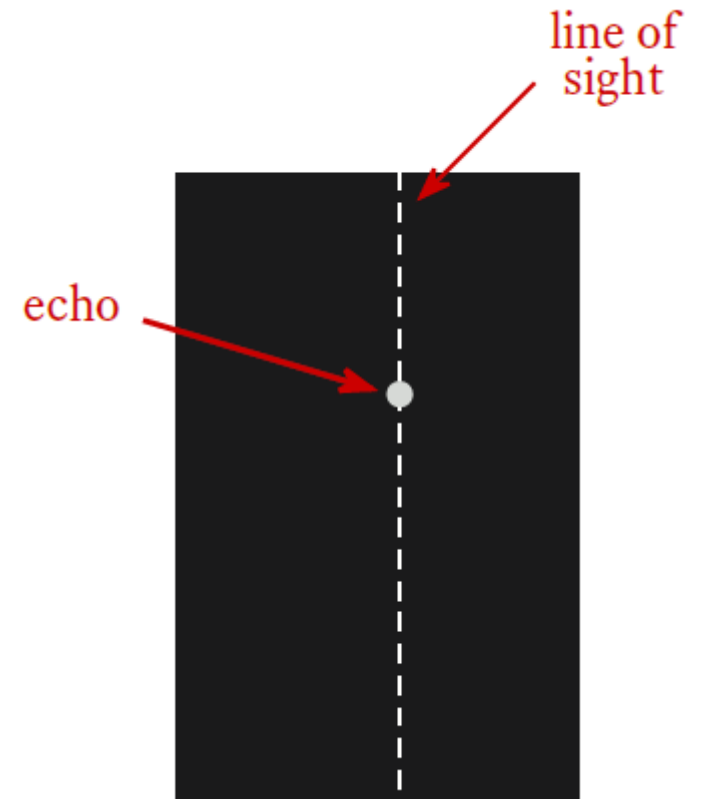
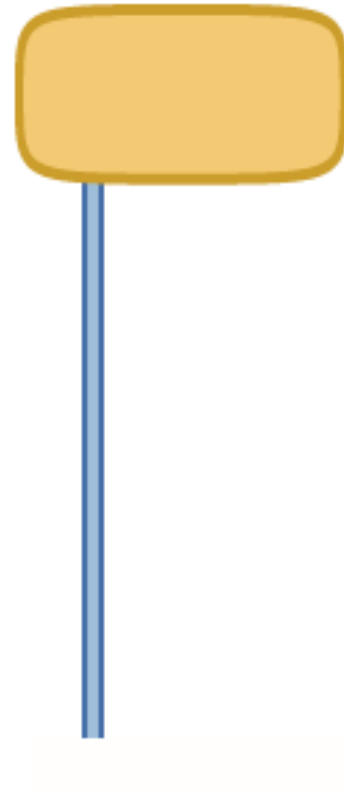
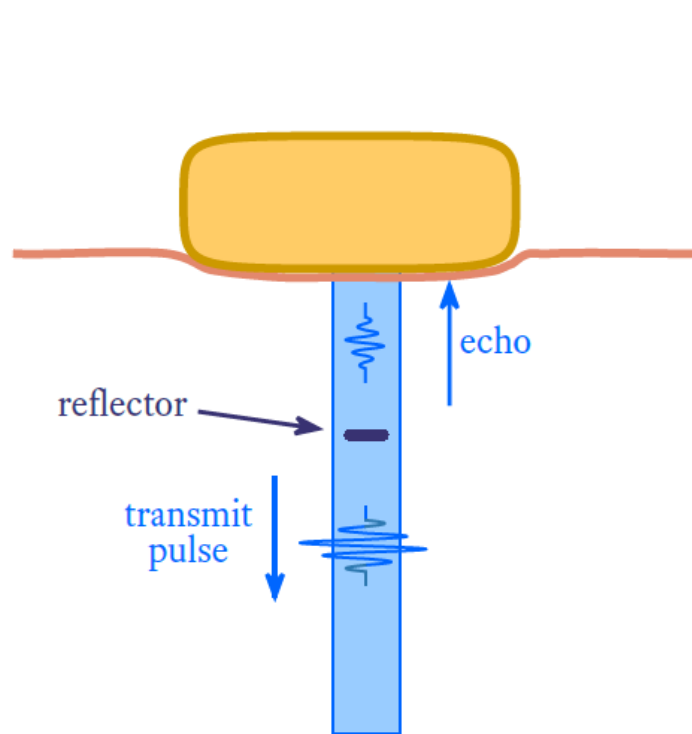
Imaging with Pulsed Ultrasound



Imaging with Pulsed Ultrasound



Imaging with Pulsed Ultrasound



Field of View

Imaging with Pulsed Ultrasound

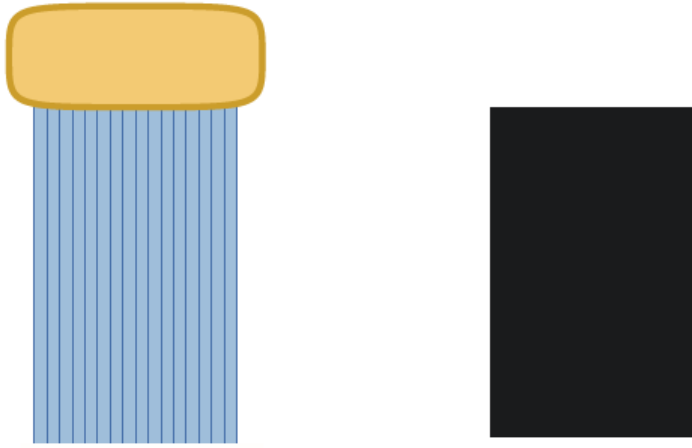


Figure 3.8 As the beam moves from one end of the probe to the other it sweeps out a rectangular field of view and so the image (black) is rectangular.

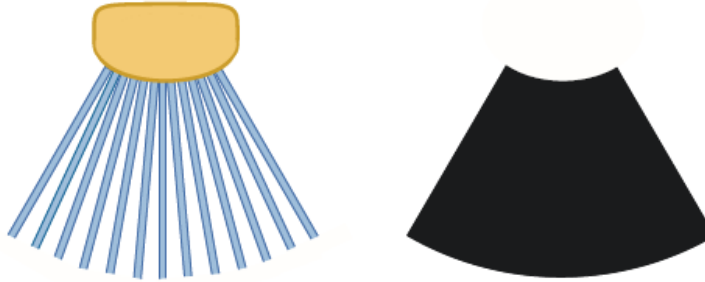


Figure 3.9 A curved array is similar to a linear array. However, due to the curvature of the probe the lines of sight are not parallel, but instead sweep out a radial path. This type of probe therefore provides a significantly wider field of view than the linear array.

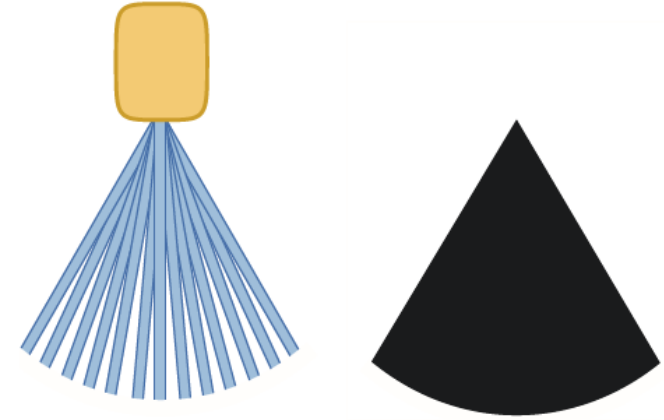
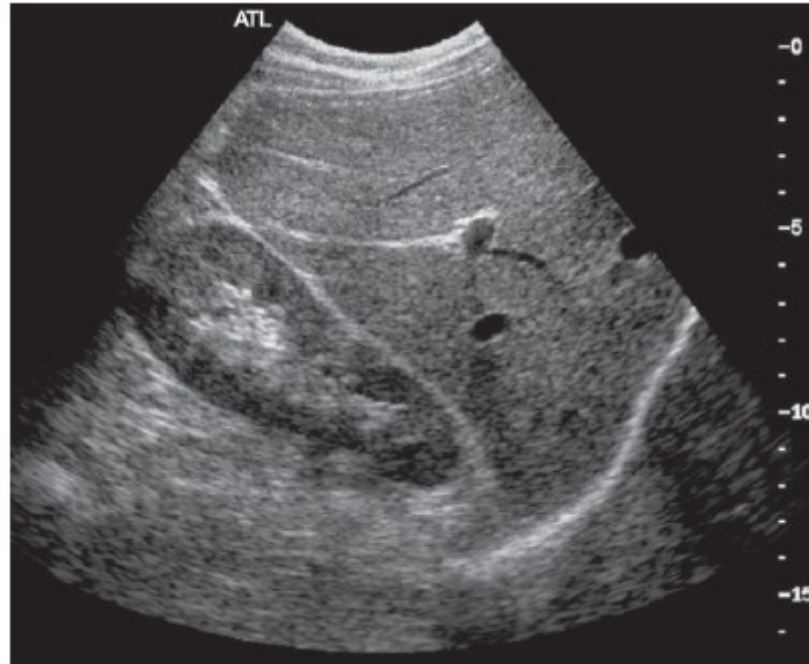


Figure 3.10 Phased array. The point of origin of the beam remains fixed. It is scanned by steering it in a series of different directions. The resulting image provides a good field of view at depth, but virtually no information about superficial tissues.

Imaging with Pulsed Ultrasound



Example of B-Mode image

B - for Brightness

2D Spatial image

Imaging with Pulsed Ultrasound

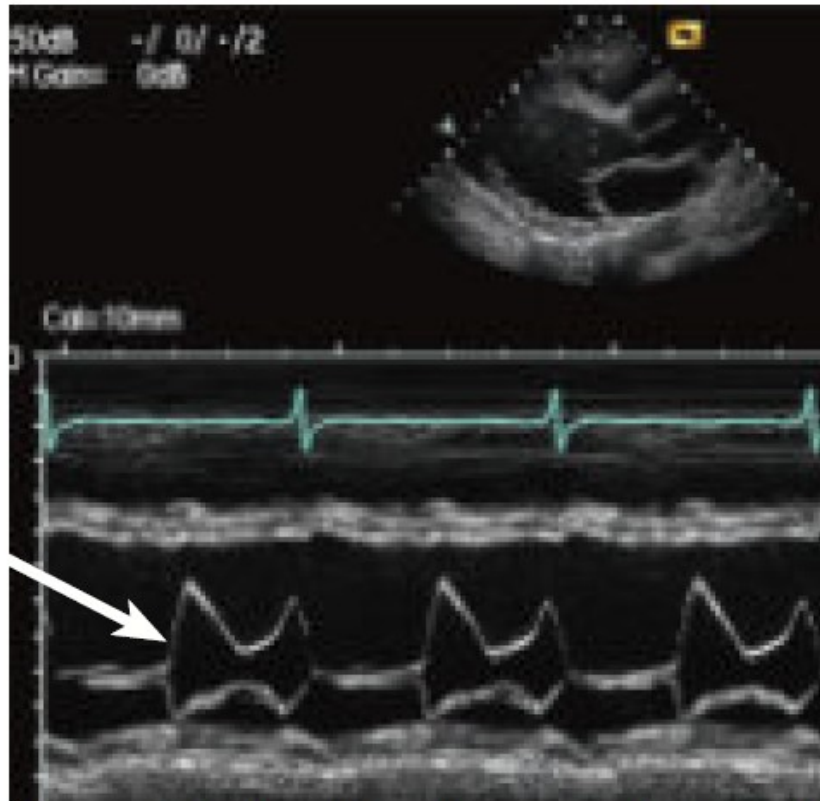
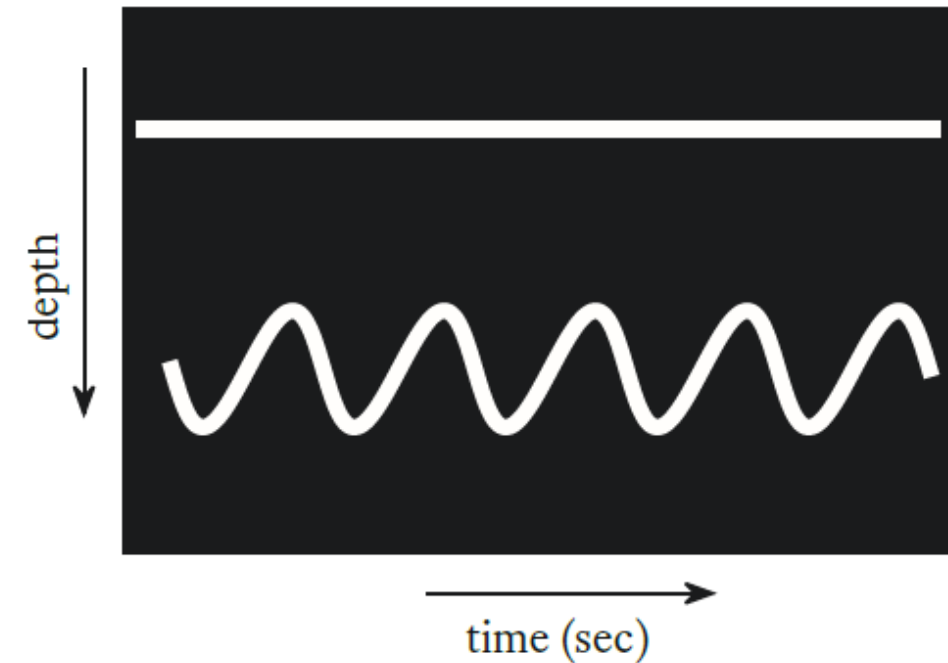
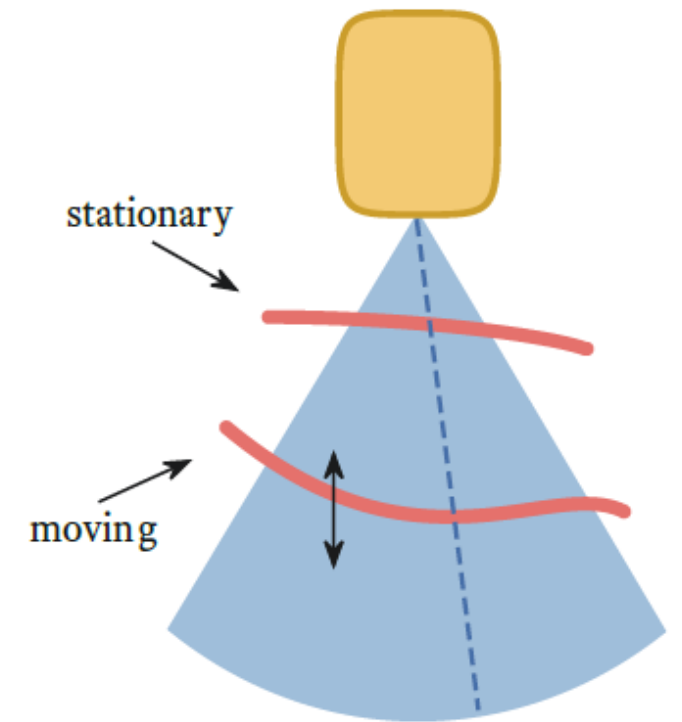


Figure 3.13. An M mode trace showing the movement of the mitral valve leaflets (arrow).

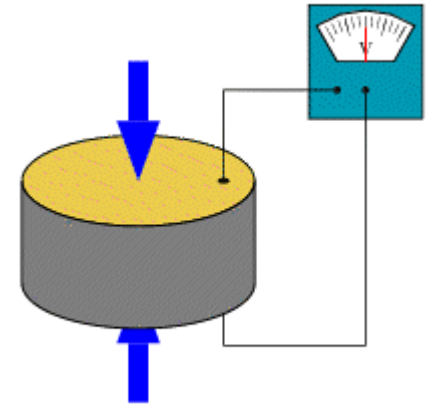
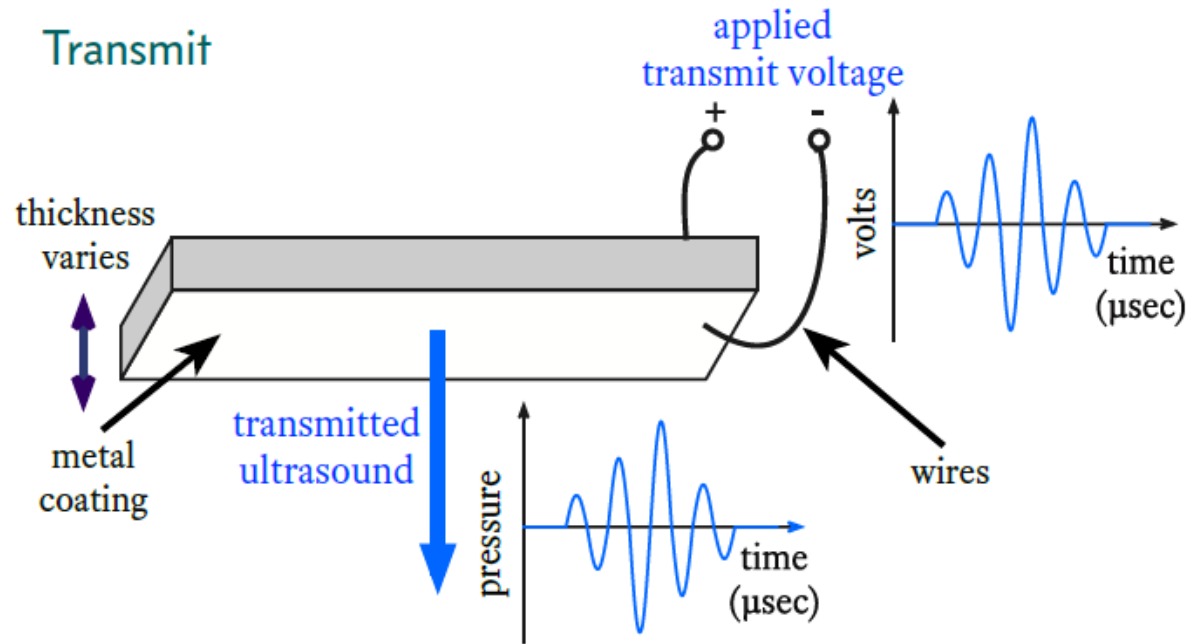
Example of M-Mode

M - for Motion

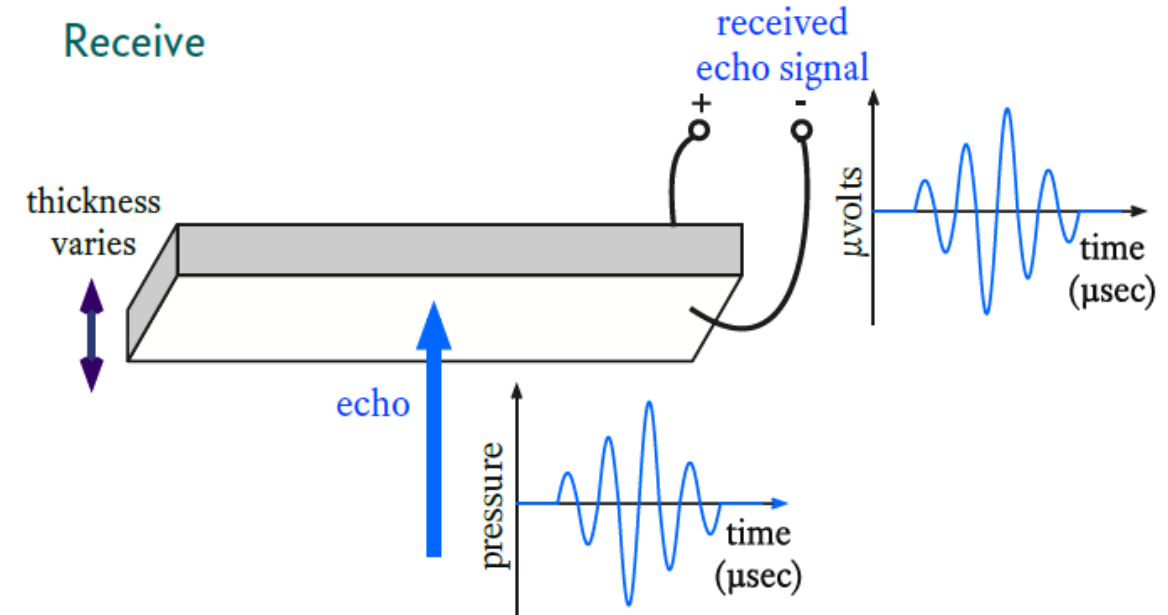


Ultrasound Transducers and Beam focusing

Transmit



Receive



Ultrasound Transducers and Beam focusing

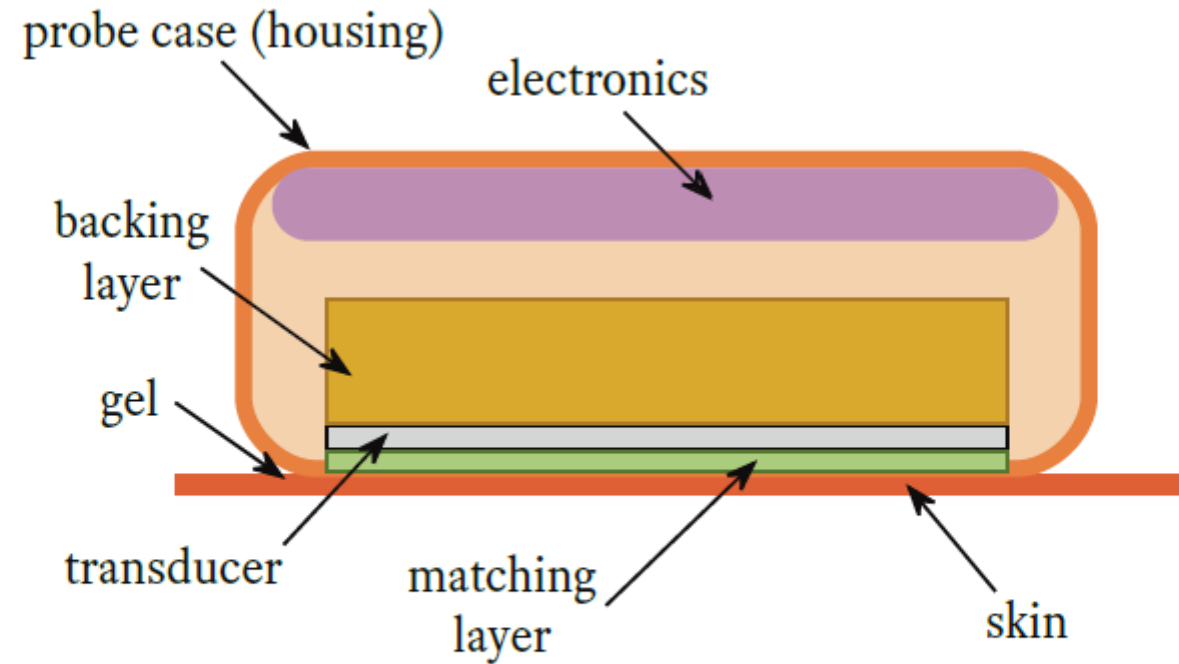


Figure 4.3 Cross-section through a linear probe showing the transducer, matching and backing layers. Also shown are the electronics that are normally found in the probe.

Ultrasound Transducers and Beam focusing

- US beams from a transducer have a finite width, based on the transducer size

Key technical specs of an ultrasound transducer for medical applications

- Bandwidth

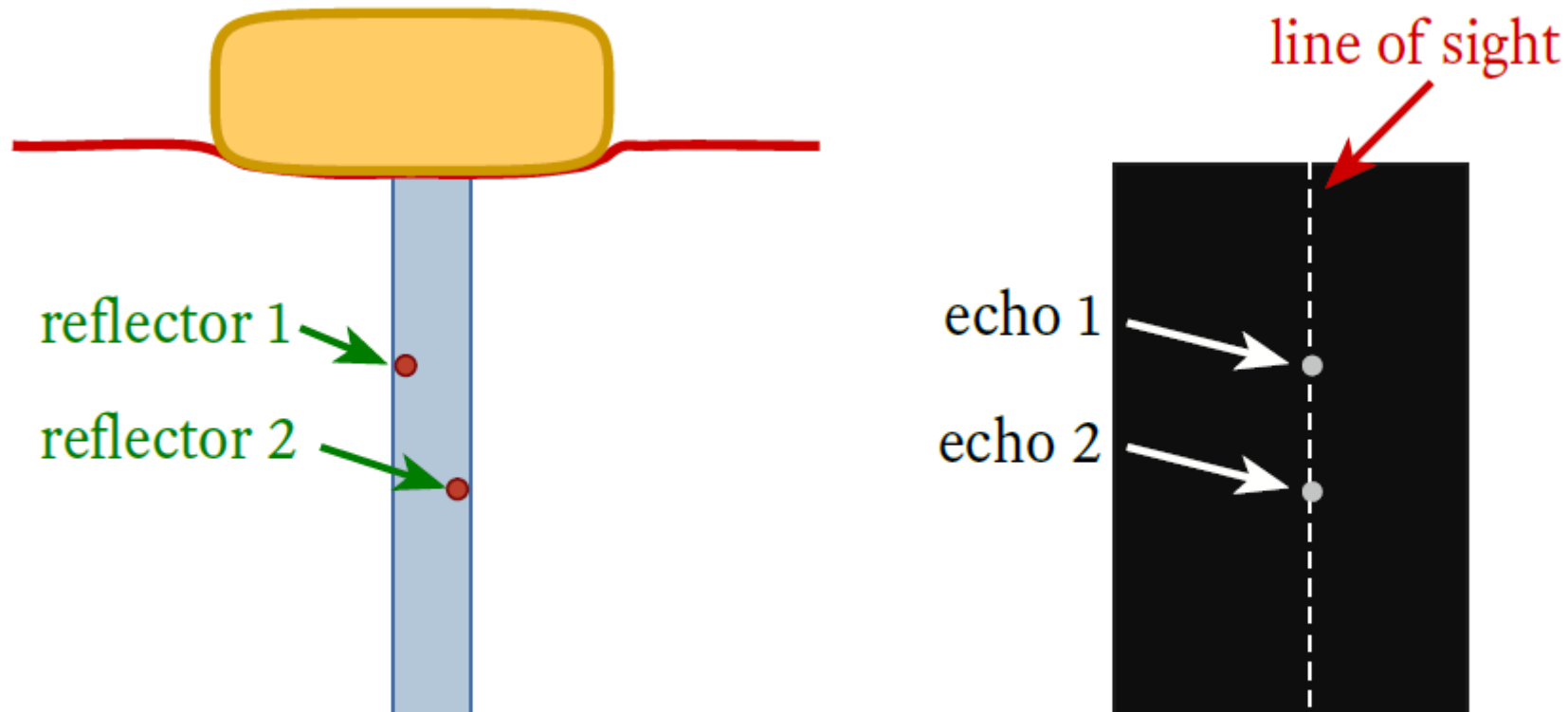


Figure 4.4 The machine places all echoes gathered by a particular beam along a line corresponding to the midline of the beam (often called the line of sight).

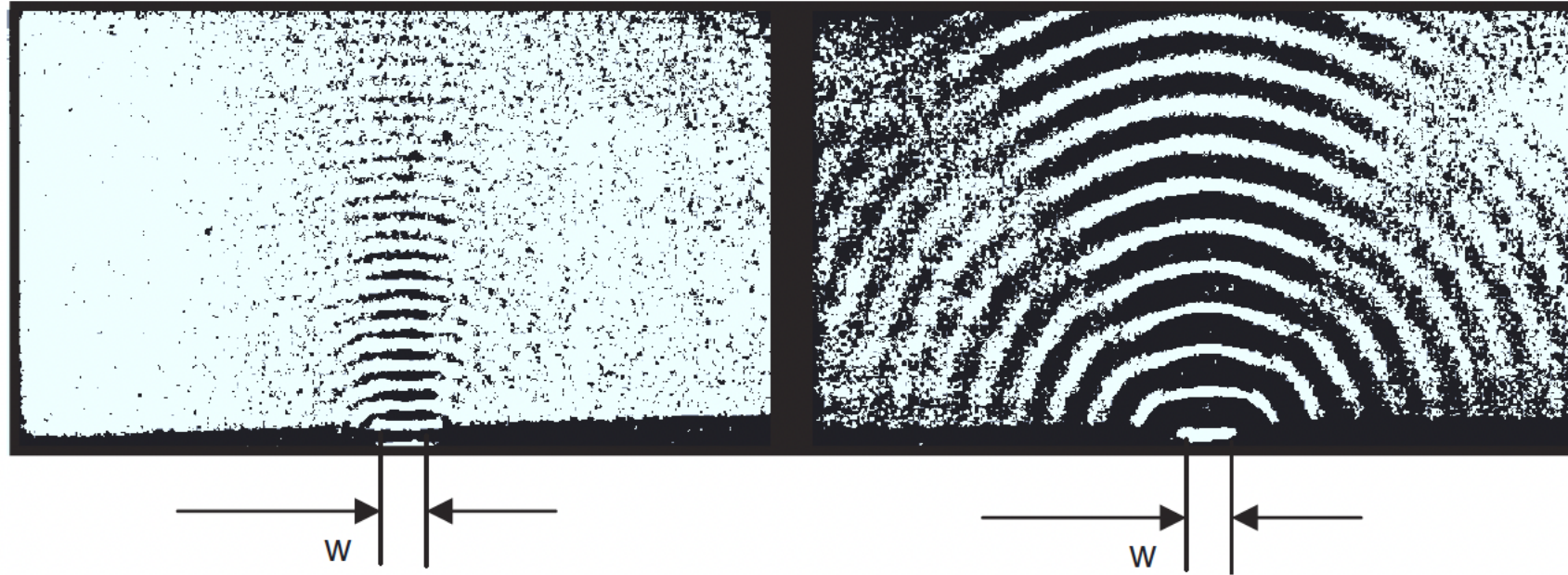
Diffraction of a plane wave through a slit

$$W = 10\lambda$$

$$\lambda/w = 0.1$$

$$\lambda/w = 0.6$$

$$W = \frac{10\lambda}{6}$$



For a perfectly parallel beam (i.e., without divergence, the aperture needs to be much larger than the wavelength of the ultrasound wave. For example, this means that for a wave with λ of 0.3mm, for obtaining a near parallel beam, aperture needs $\sim 30\lambda = 9\text{mm}$, just for 1 beam. This would severely limit our ability to create an image with reasonable resolution.

Practically....the ultrasound transducers' pitch (width) in a transducer array is typically on the order of a few 10s to 100s of micro-metres, depending on the frequency range supported by the transducer.

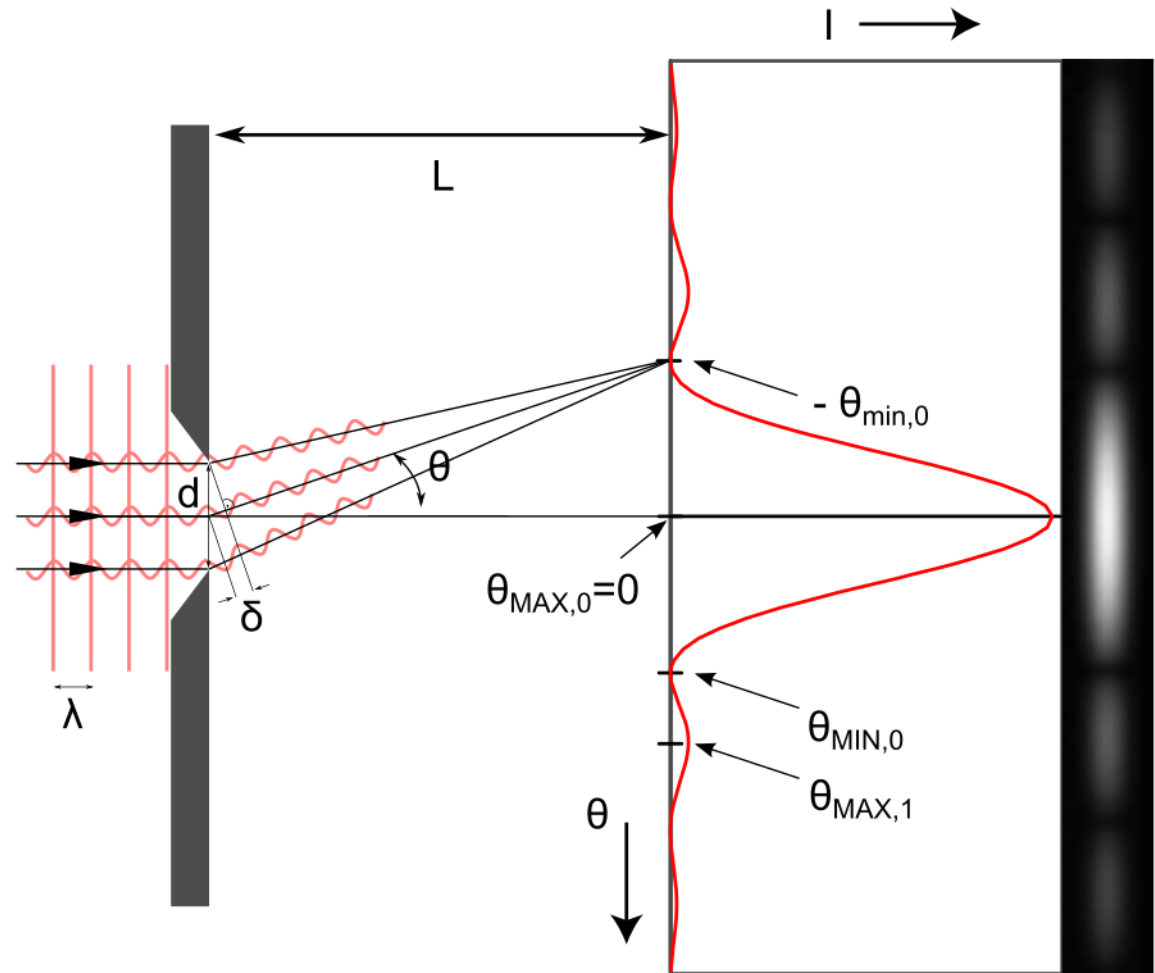
Diffraction of a plane wave through a slit

Diffraction example from Light propagation

The difference in path length determines the divergence angle, which in turn is dependent on the relative amplitudes of the aperture and the wavelength

In case of light propagation

$$\sin \theta = \frac{\lambda}{d}$$



Ultrasound Transducers and Beam focusing

- US beams from a transducer have a finite width, based on the transducer size

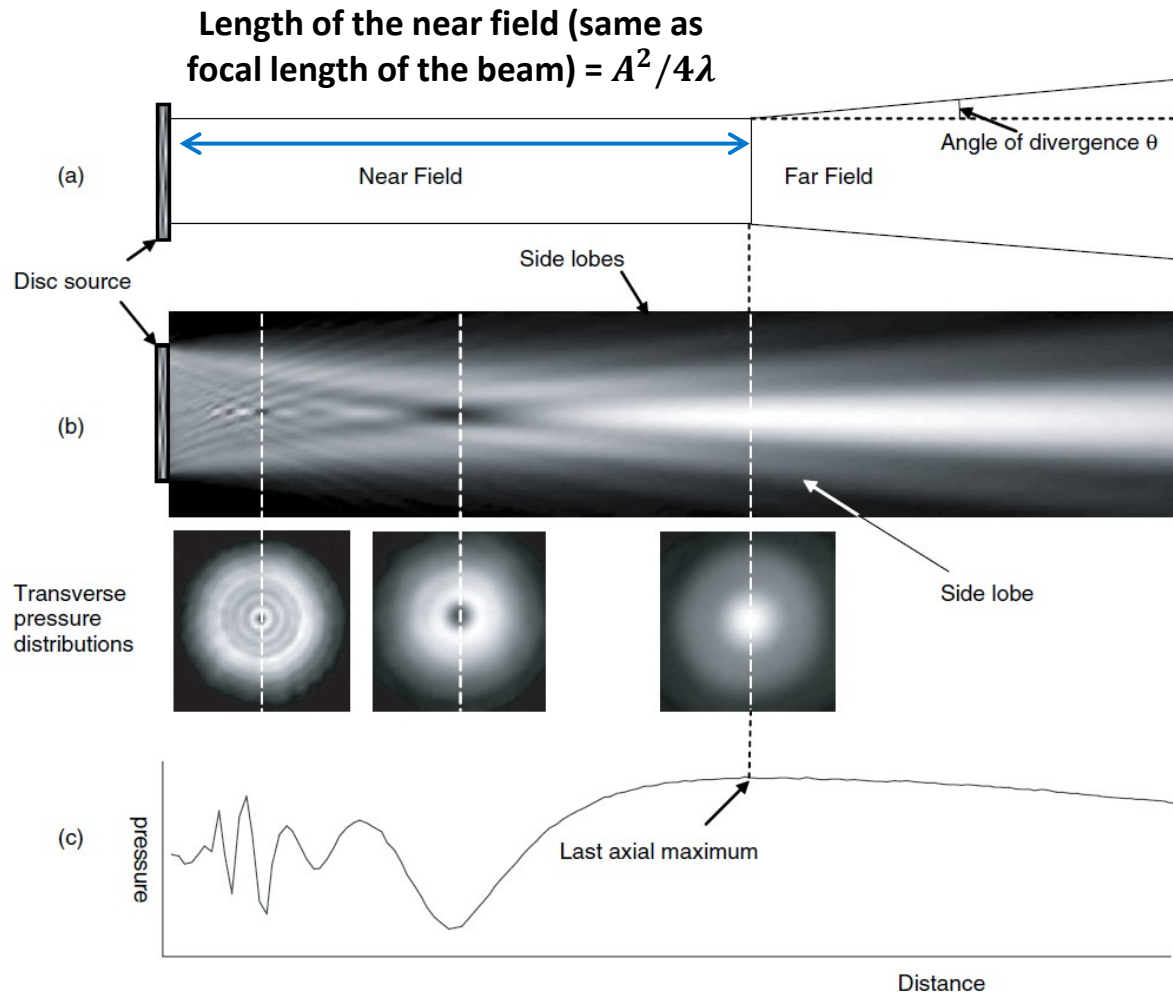
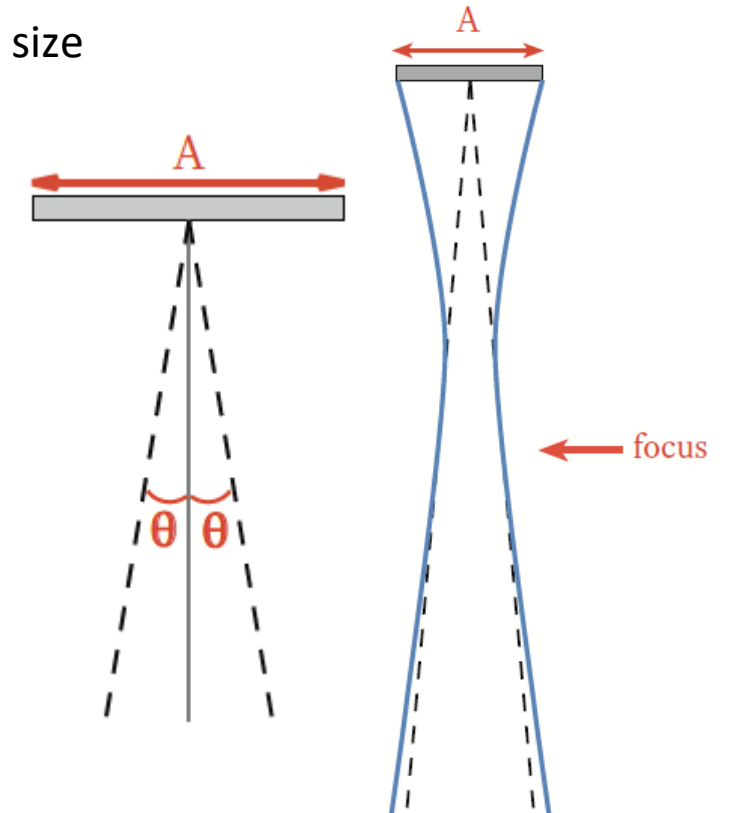


Fig. 2.15 The ultrasound beam from a plane disc source consists of a near field, in which the pressure distribution is complex, and a far field, in which it is more uniform.



$$\sin \theta = \frac{1.22\lambda}{A}$$

$$\theta = \sin^{-1} \left(\frac{1.22\lambda}{A} \right)$$

Figure 4.5 A focussed ultrasound beam showing the diffraction limit (broken lines).

For circular aperture
with diameter of A

Ultrasound Transducers and Beam focusing

- US beams from a transducer have a finite width, based on the transducer size

$$\sin \theta = \frac{1.22\lambda}{A}$$
$$\theta = \sin^{-1} \left(\frac{1.22\lambda}{A} \right)$$

$$\begin{aligned} \text{beam width} &= 2 \tan \theta F \\ &= 2 \left(\frac{\sin \theta}{\cos \theta} \right) F \end{aligned}$$

$$\begin{aligned} \text{beam width} &= 2 \sin \theta F \\ &= 2 \left(\frac{1.22\lambda}{A} \right) F \\ &= \frac{2.44\lambda F}{A} \end{aligned}$$

Considering F is $\frac{A^2}{4\lambda}$, **beam-width** $\approx \frac{A}{2}$

Limit of the lateral resolution is given by the beam-width

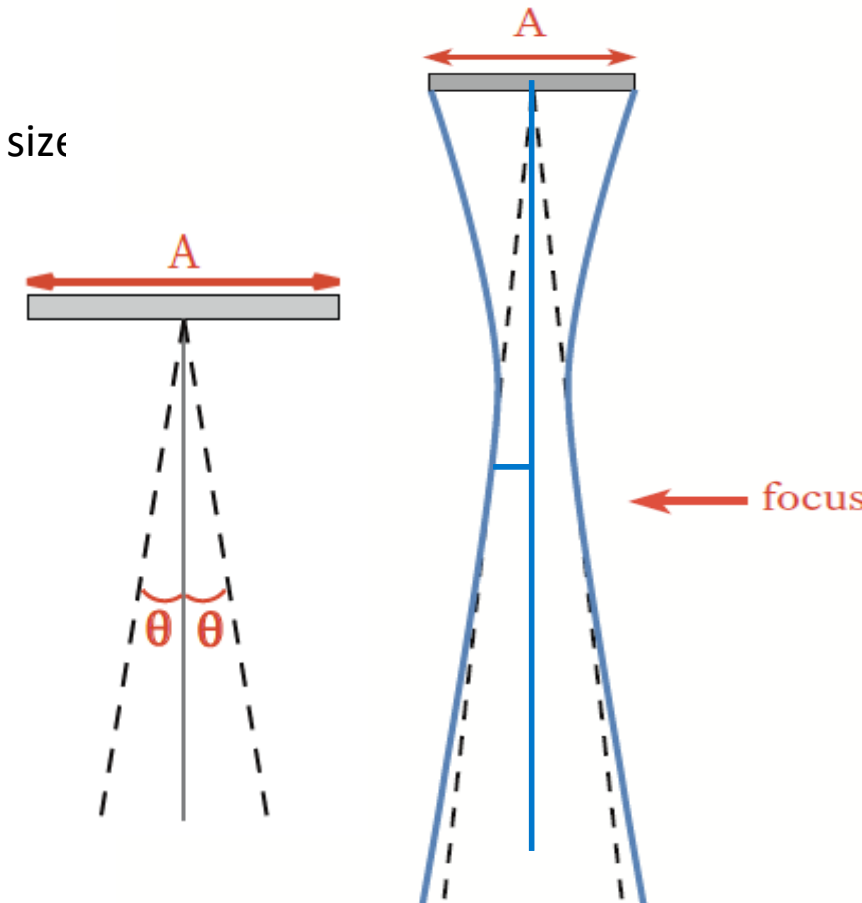
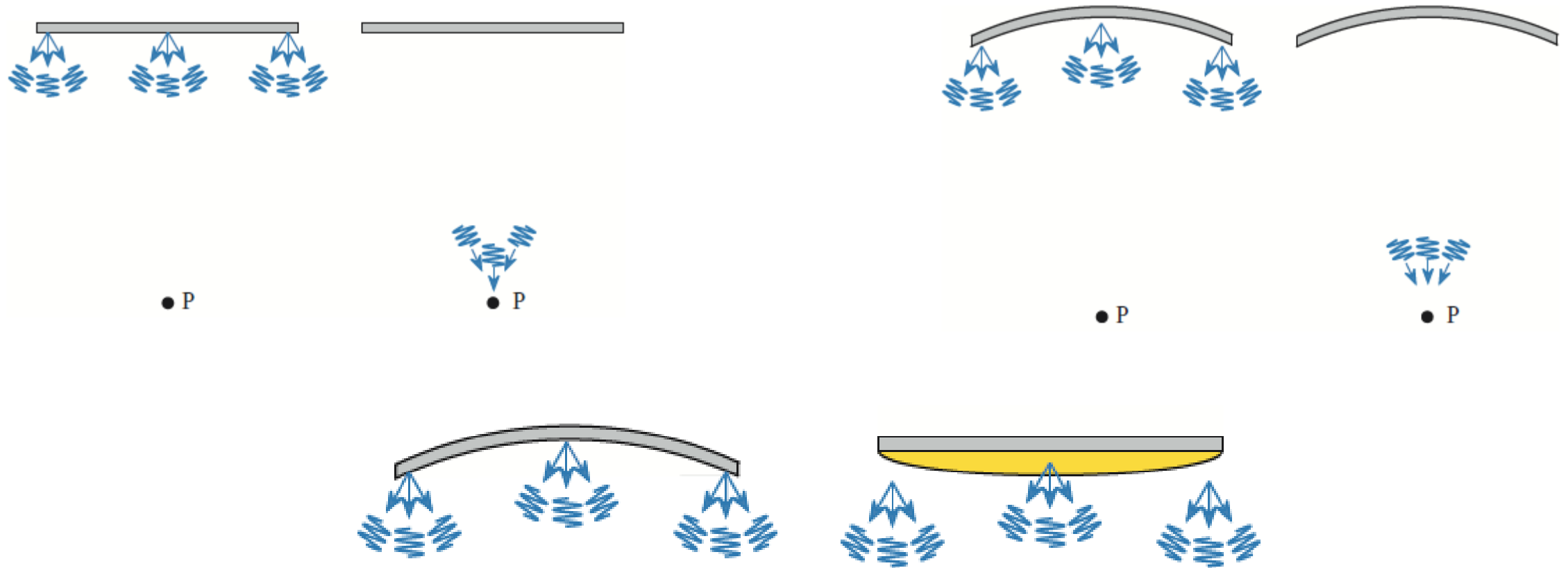


Figure 4.5 A focussed ultrasound beam illustrating the diffraction limit (broken lines,

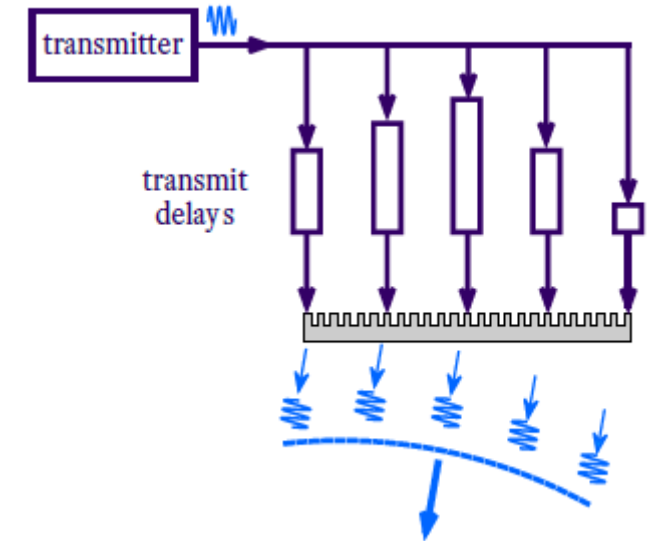
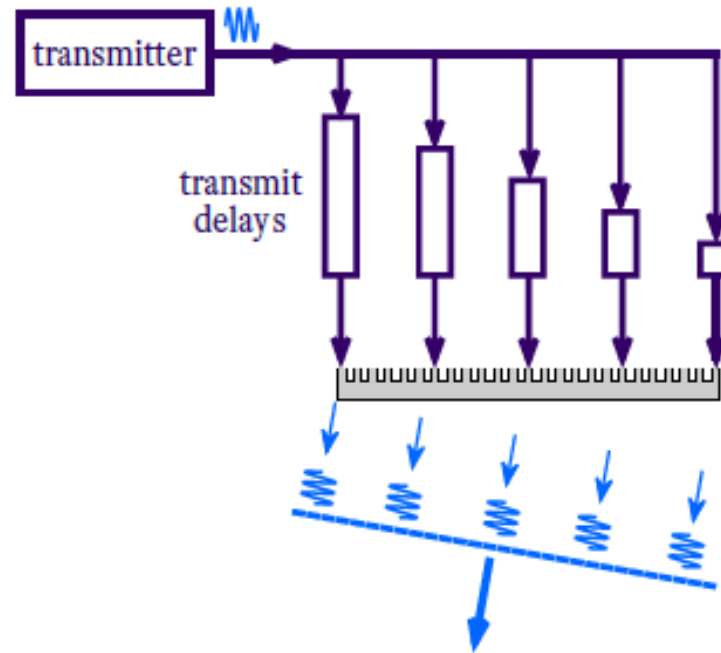
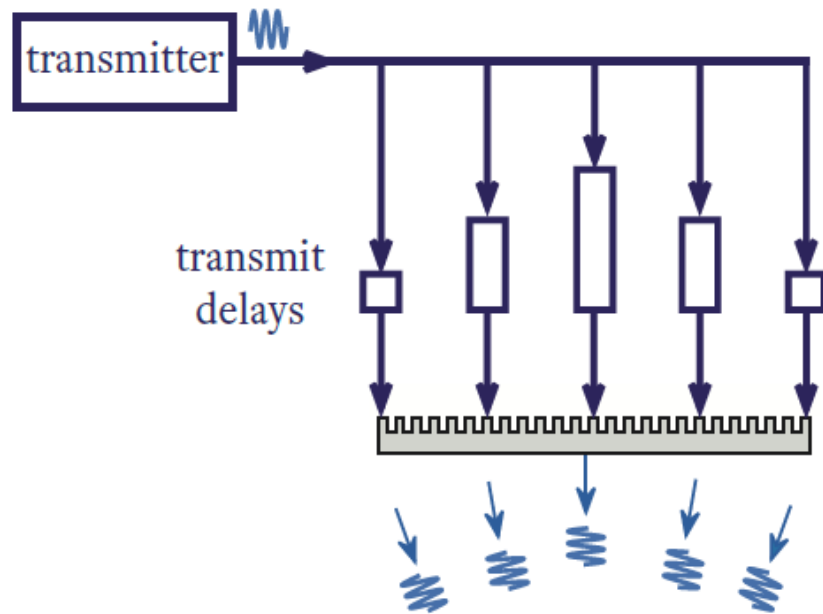
For circular aperture
with diameter of A

Ultrasound Transducers and Beam focusing



Focusing lens

Ultrasound Transducers and Beam focusing and steering



Ultrasound Transducers and Beam focusing and steering

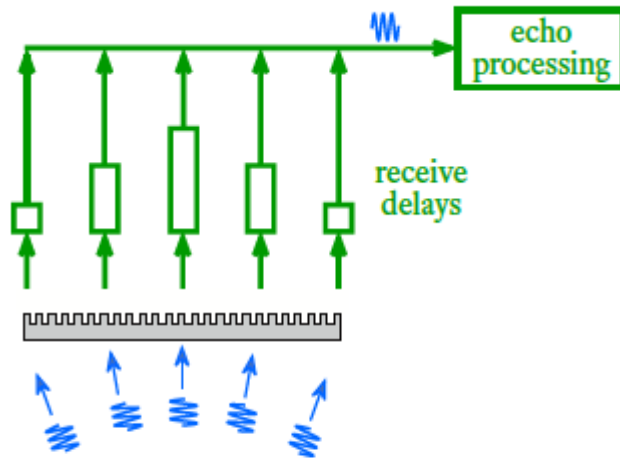
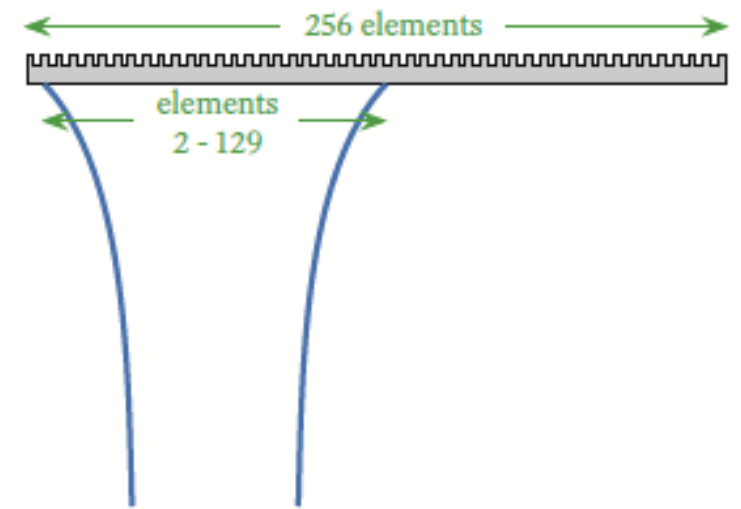
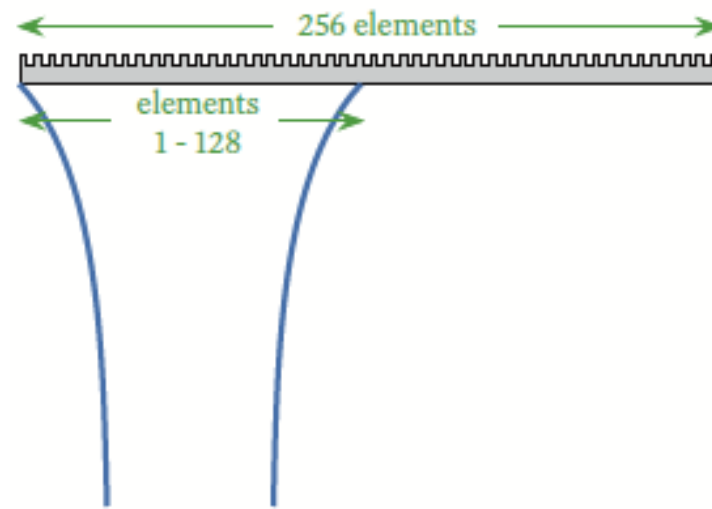
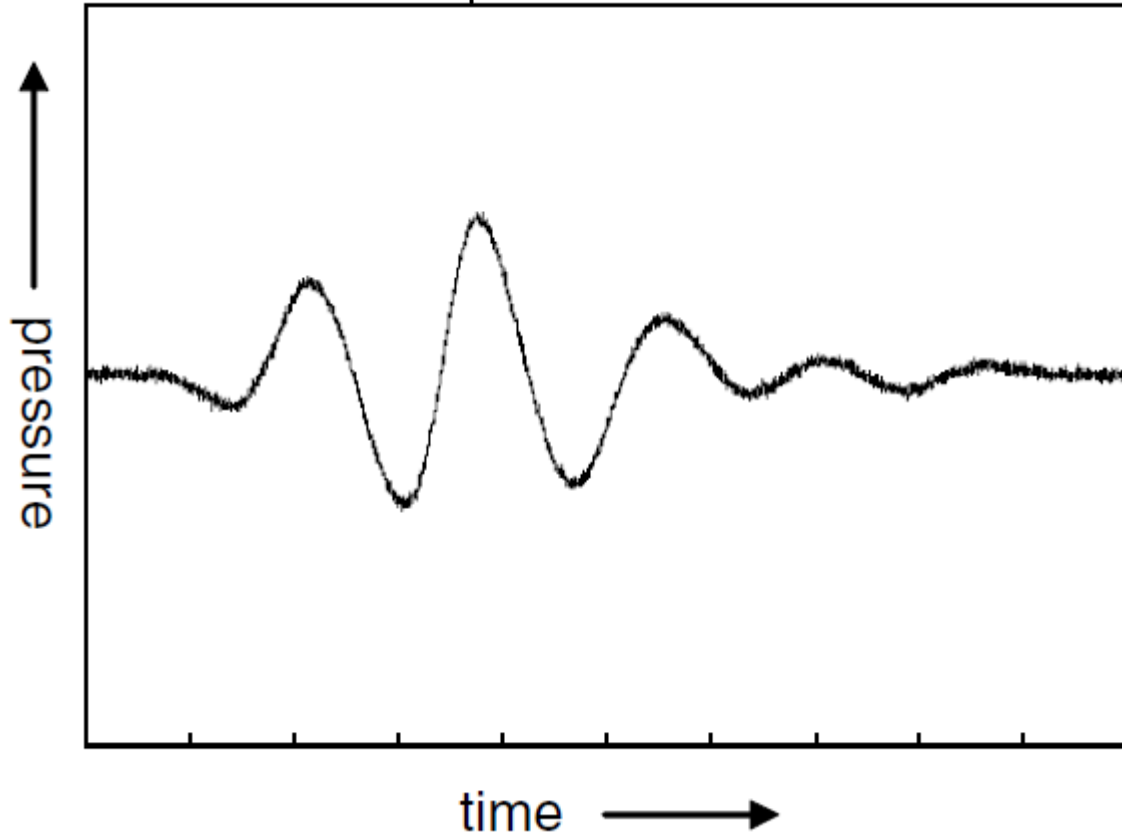


Figure 4.32 Receive focussing. The echo arrives at the centre element first, so it requires the longest delay. Conversely, the echo arrives at the outermost elements last and so they require the shortest delay.

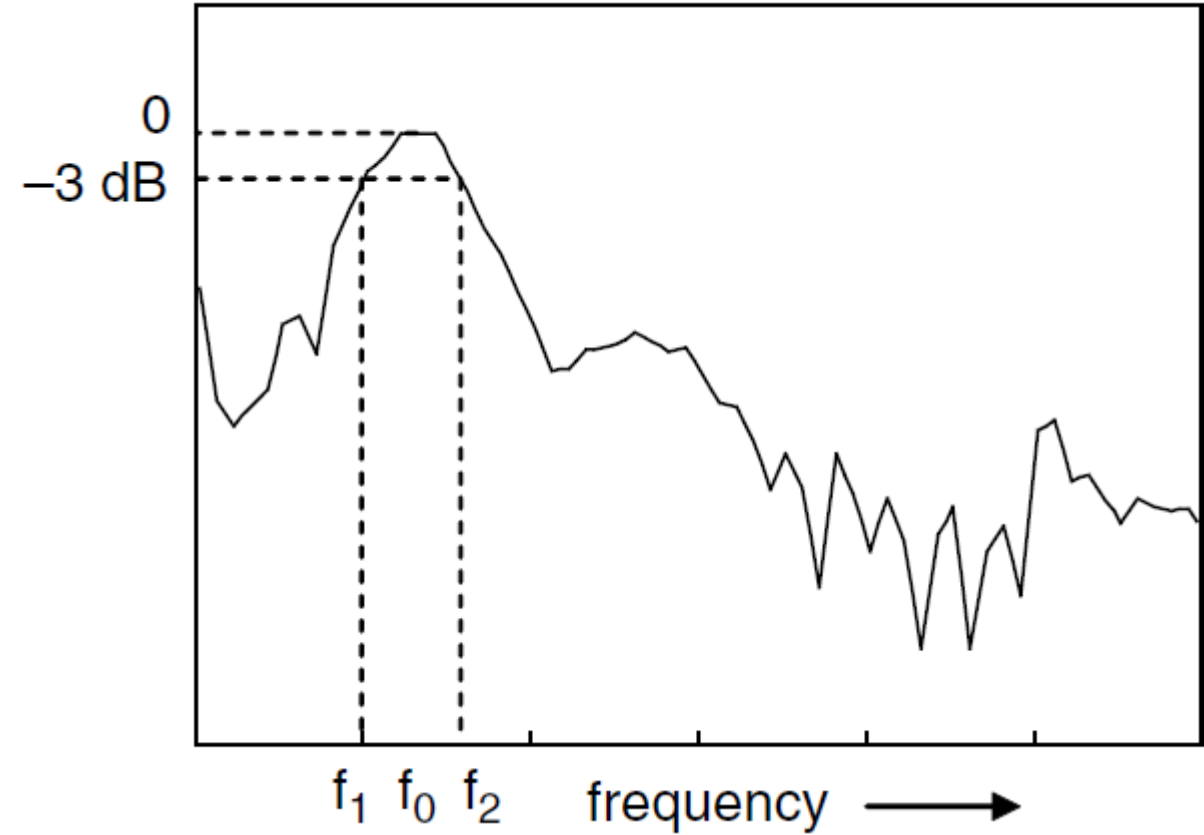


Ultrasound Pulse Characteristics

Ultrasound pulse in time domain

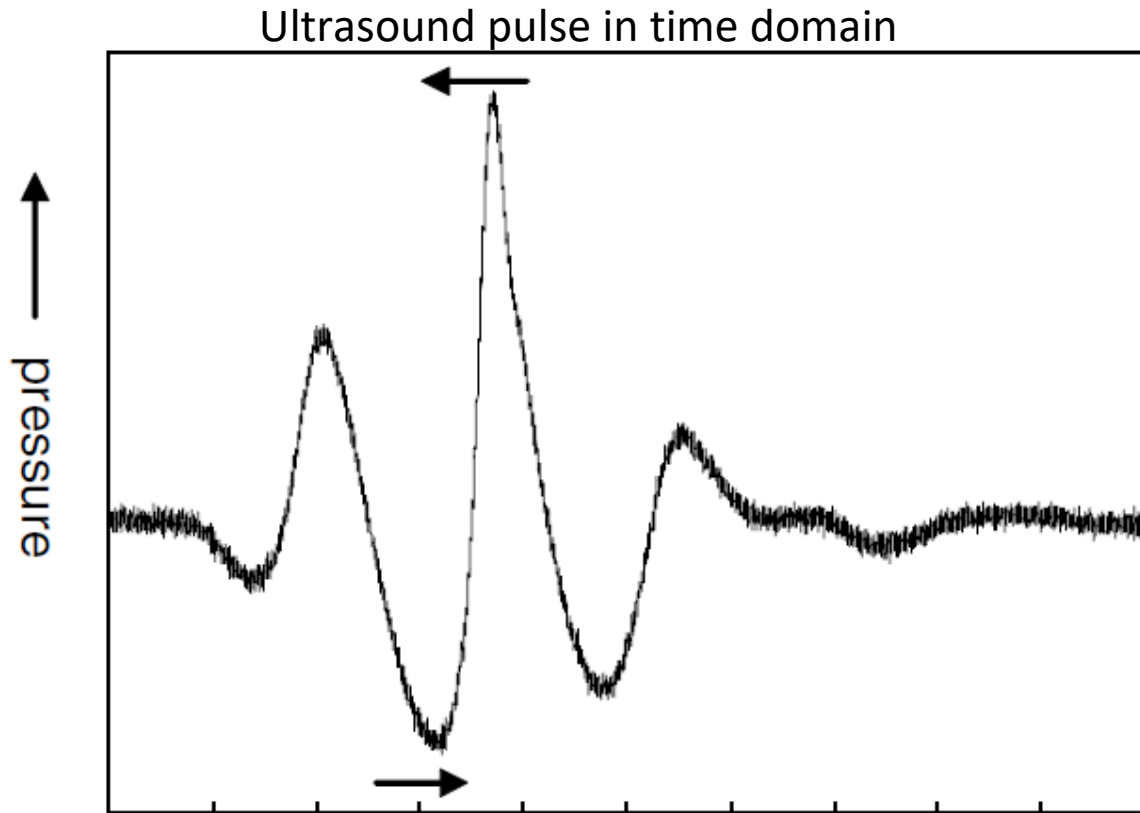


Pulse Spectrum: Ultrasound pulse in frequency domain

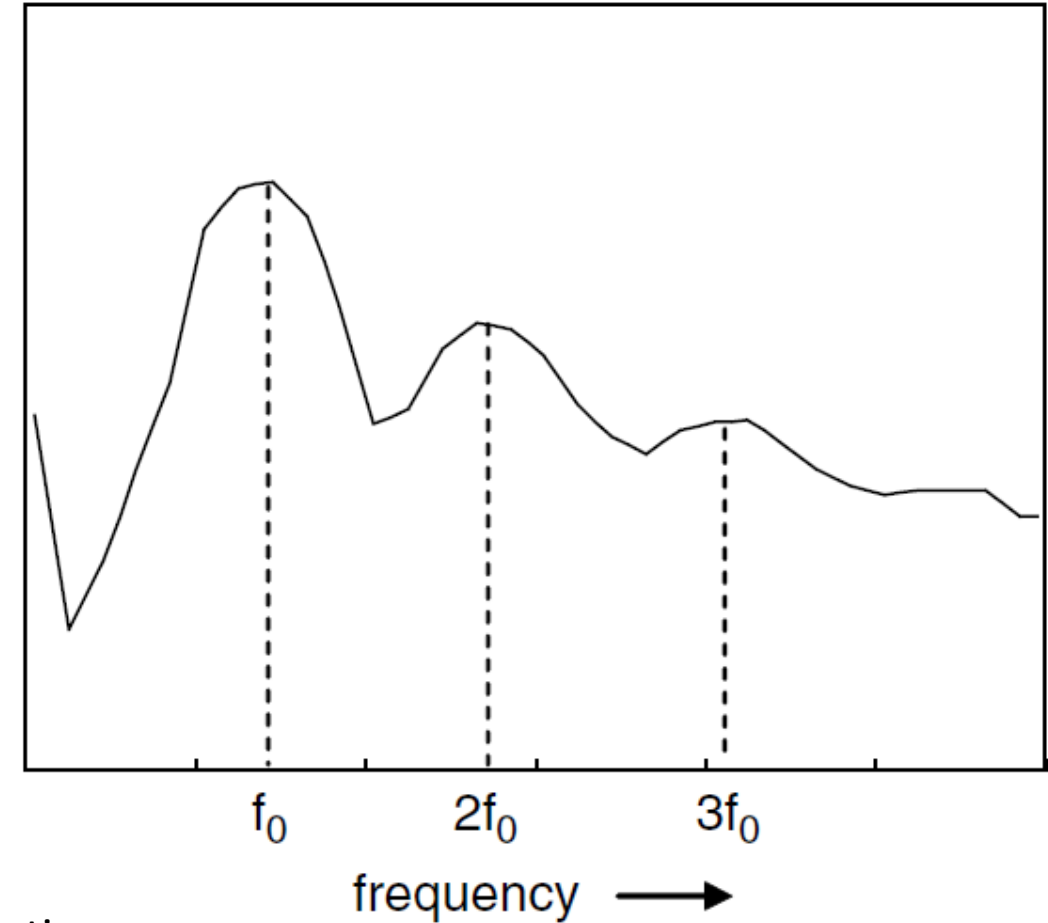


An ultrasound pulse contains many frequencies other than the intended center frequency and the pulse duration determines the frequency content

Ultrasound Pulse Characteristics

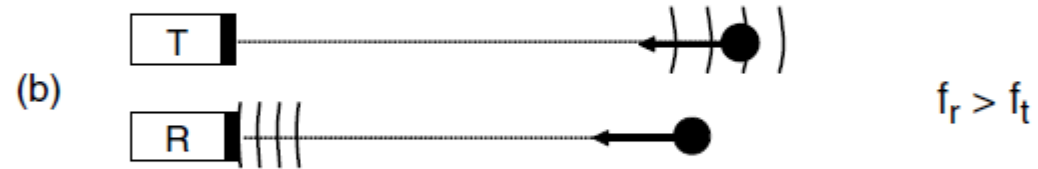


Pulse Spectrum: Ultrasound pulse in frequency domain



- Longer spatial pulse length adversely affects axial resolution.
- However, since the pulse bandwidth is narrower (energy more concentrated around the center frequency), it has better penetration depth and also higher signal to noise ratio

Doppler imaging (Doppler shift)



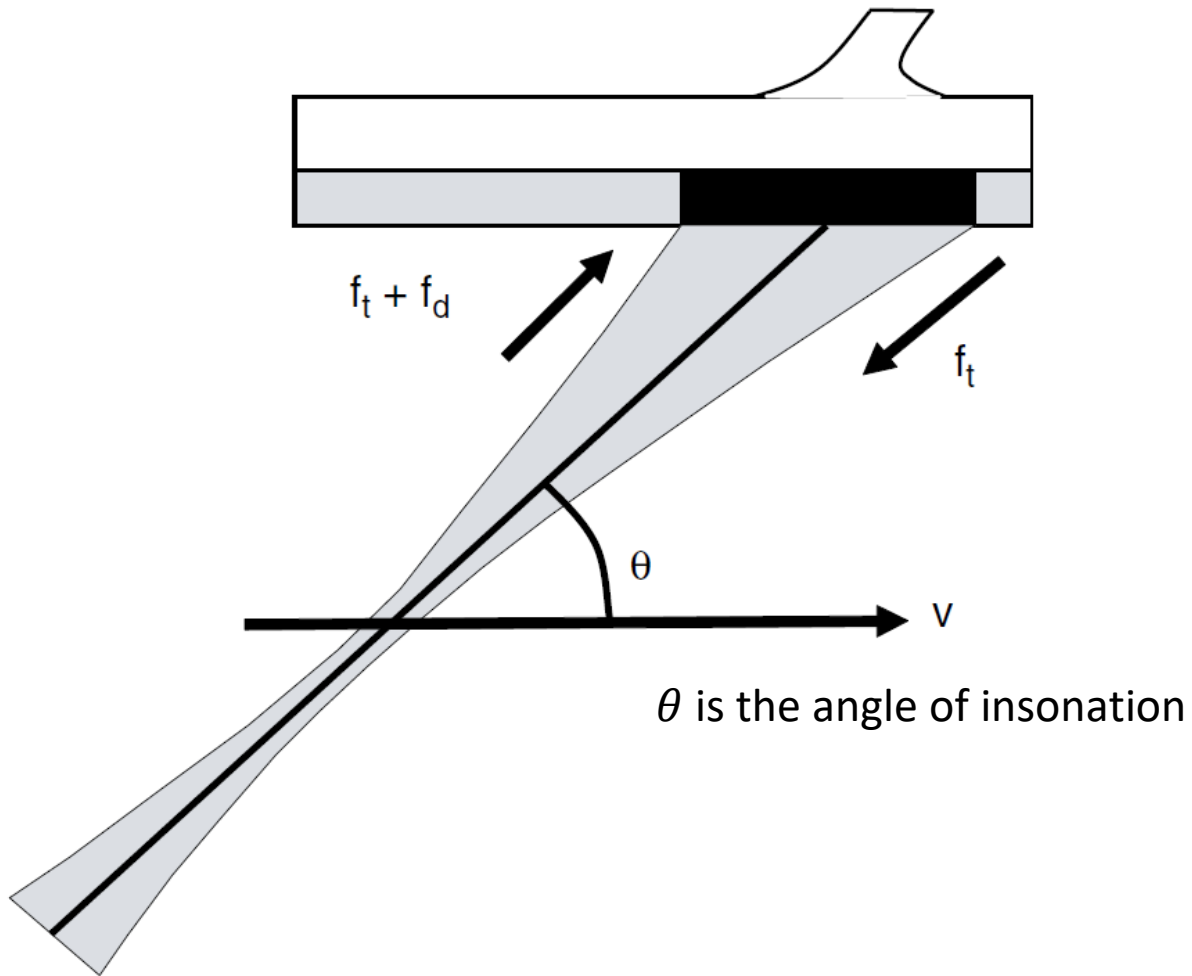
$$f_o = f_t \frac{(v \pm v_o)}{(v \mp v_s)}$$

- Source is stationary
- Reflector is moving

$$v_s = c \cdot \frac{(f_o - f_t)}{(f_o + f_t)}$$

$$v_s \approx c \cdot \frac{(f_o - f_t)}{2f_t}$$

Doppler imaging (Doppler shift)



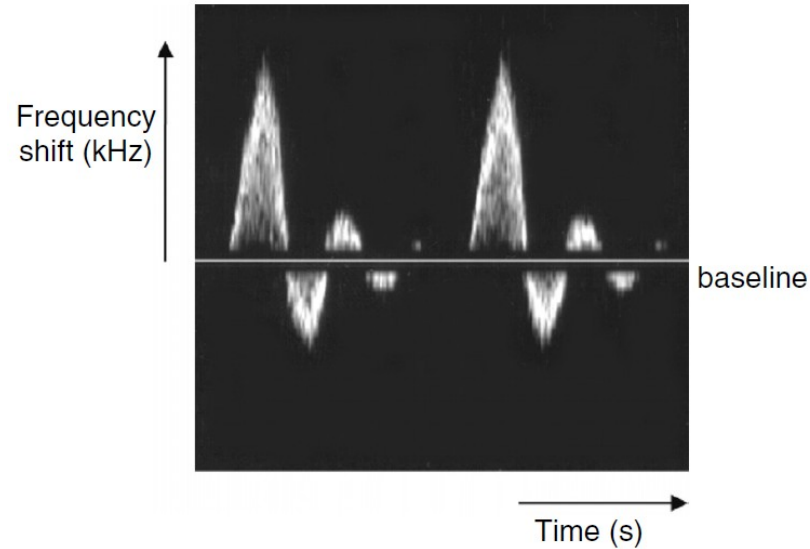
$$v_s \cdot \cos \theta = c \cdot \frac{(f_o - f_t)}{2f_t}$$

Doppler shift detected

$$f_d = f_r - f_t = \frac{2f_t v \cos \theta}{c}$$

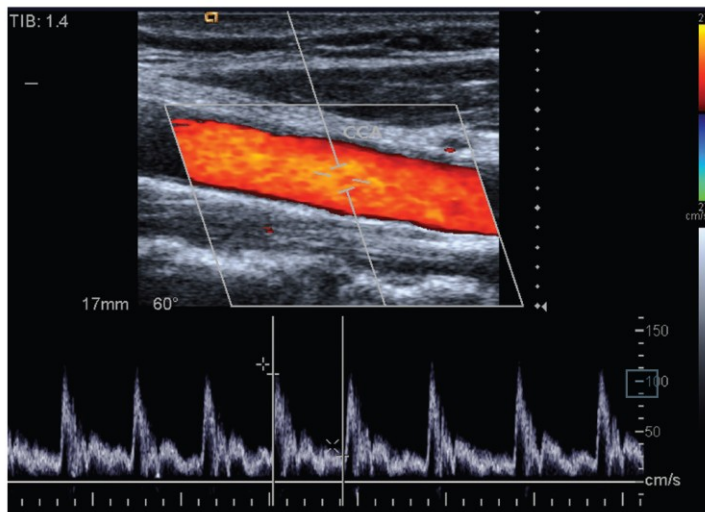
$$v = \frac{c f_d}{2f_t \cos \theta}$$

Doppler imaging (Doppler shift)



Spectral Doppler

- Deviation from the baseline indicates the amount of Doppler shift
-
- The brightness indicates the echo amplitude at that shift

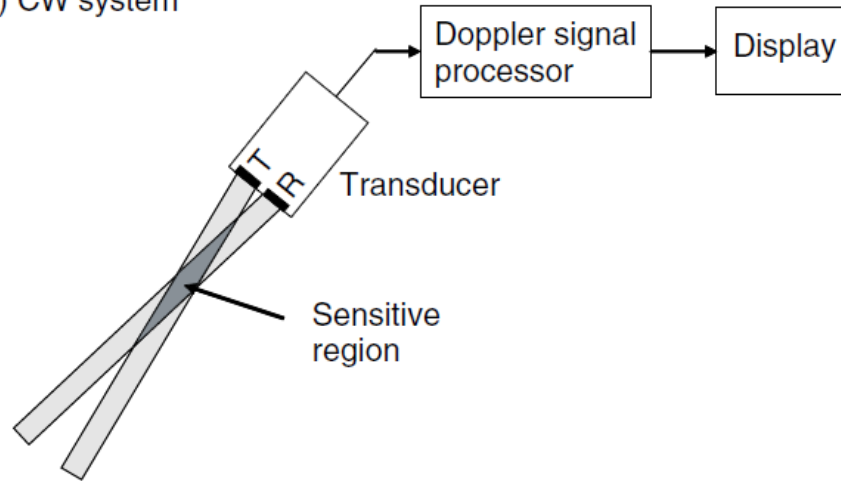


2D Color Doppler

- The color represents Doppler shift from that location
- Allows us to examine velocity changes in a spatially resolved manner

Continuous Wave vs Pulse Wave Doppler

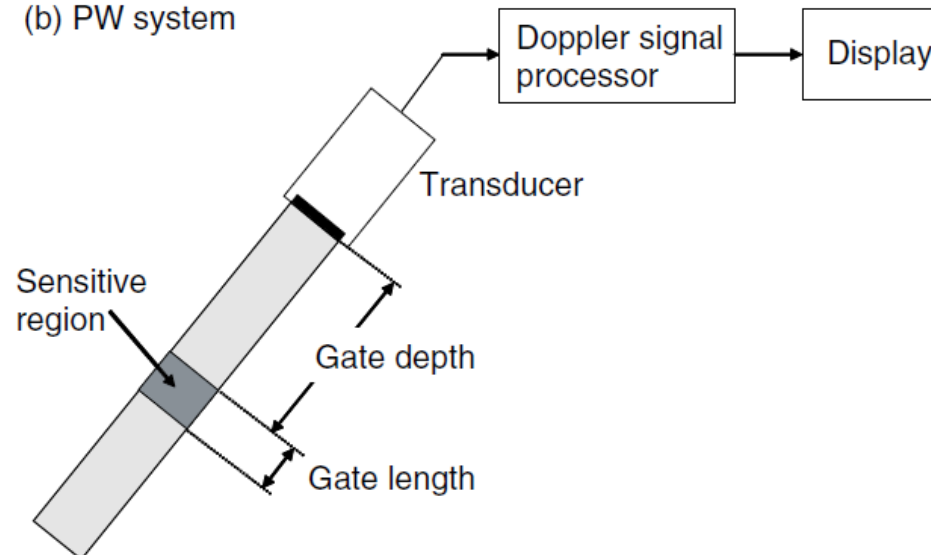
(a) CW system



Continuous Wave Doppler

- Separate transmit and receive transducers
- Intersection of the transmit and receive beam paths determine the sensitive region

(b) PW system



Pulse Wave Doppler

- Single element used for both transmit and receive
- 'Gate' region, a user-controlled parameter, determines the sensitive region

Continuous Wave vs Pulse Wave Doppler

Doppler Signal Processing

- Need to separate the blood signal from the echoes arising from tissue (moving and stationary)
- Discerning parameters – echo amplitude vs. Doppler-Shift
- Echoes from moving blood have high Doppler-shift but low amplitude
- Echoes from the moving tissue is converse to this.

	Velocity ranges	Signal intensity
Blood	0–600 cm s ⁻¹	Low
Tissue	0–10 cm s ⁻¹	40 dB higher than blood

Biosafety Effects of Ultrasound imaging

As the Ultrasound pulse travels through the tissue, it gets attenuated. The lost intensity is absorbed by the body.

- Mechanical effects
- Thermal effects

Ultrasound Factors affecting temperature increase (tissue heating)

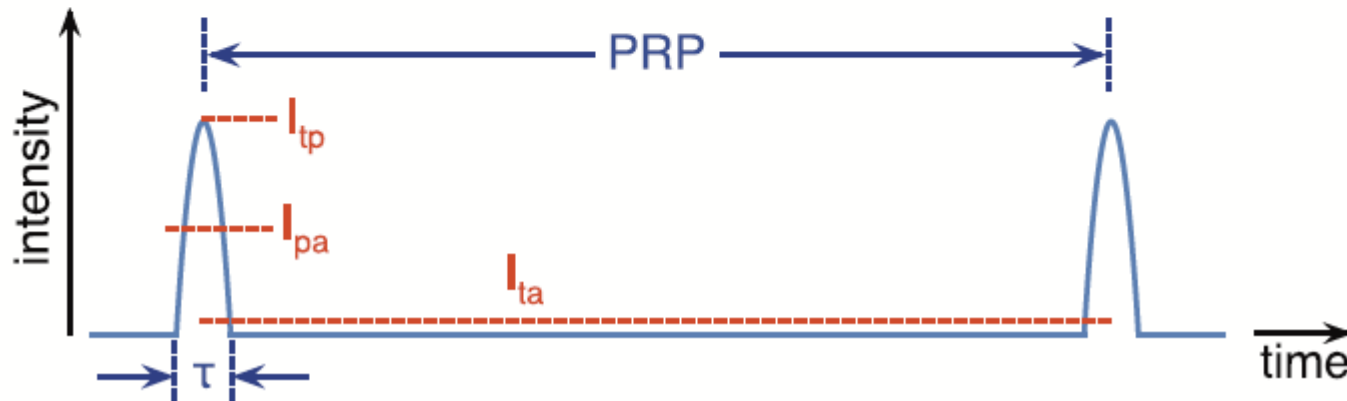
- Intensity
- Frequency
- Ultrasound absorption properties of tissues
- Ultrasound conduction property
- Blood perfusion through the tissue (heat sink)

Biosafety Effects of Ultrasound imaging



$$I_{pa} = \frac{P_p}{\tau}$$

- P_p is the energy per square centimetre delivered during one transmit pulse
- I_{pa} is the Pulse average intensity



$$I_{ta} = \frac{P_p}{PRP}$$

- I_{ta} is the temporal average intensity delivered over a pulse repetition period

Biosafety Effects of Ultrasound imaging

$$DF = \frac{\tau}{PRP} = \frac{I_{ta}}{I_{pa}}$$

- DF – Duty Factor
- τ – Transmit Pulse duration
- PRP – pulse repetition period

Typical DF value ranges from 0.1% to 1%.

What is important is to avoid temperature increase

- Based on the anatomy being scanned, and the frequency being used, an estimate of the transmitted ultrasound power (W_{deg}), that would lead to a 1 degree increase.

Tissue specific Thermal Index, TI

$$TI = \frac{W}{W_{deg}}$$

- Soft tissue TI_s
- Bone TI_b
- Cerebral TI_c

